

Democratic Voter Shifts in Pennsylvania: Turnout vs. Persuasion, and the Roadblocks of Testing Text-Based Explanations at the County-Level

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Abstract—This review synthesizes research on (1) how county-level voter turnout and partisan vote share change in U.S. elections, (2) the relative importance of mobilization versus persuasion in explaining Democratic losses,, (3) county demographics and political behavior in Pennsylvania and similar swing states, and (4) the difficulties of using open-ended survey responses and text-analysis methods (dictionaries, topic models) to explain county-level electoral outcomes. The literature supports a two-part strategy: first, define and measure "shift away from Democrats" using county-level turnout and vote-share changes; second, use text-derived issue emphasis and sentiment from open-ended responses to explain variation across counties. This foundation justifies a data-science project that links Pennsylvania election data to political-attitude text data at the county level.^{[2][3][4][5][6][7][8][9][1]}

Keywords—voter turnout, mobilization, persuasion, county-level analysis, Pennsylvania, text analysis, open-ended survey responses, dictionary method, topic modeling, data science

I. Voter Turnout and Partisan Shifts in U.S. Elections

County-level turnout and vote share are the standard units for analyzing partisan shifts in U.S. elections, measured respectively as the share of voting-age or registered voters who cast ballots and as changes in party vote share across elections. Turnout can vary dramatically across counties even in high-turnout years: Pennsylvania's overall 2024 turnout reached record levels, but county-level patterns remained highly heterogeneous, with sharp urban–rural divides. Because statewide averages can mask this local variation, county-level measures are essential for distinguishing whether a party's loss is driven by turnout collapse among its base or by voters switching to the other party.^{[1][2]}

Turnout is strongly influenced by demographic factors such as age, education, race, and rural/urban status: rural and older voters tend to turn out at higher rates but shift partisan preferences over time, while younger and minority voters show greater turnout volatility. County-level changes in turnout and vote share thus reflect both demographic composition and contextual factors like mobilization, local economic conditions, and salient issues.^{[10][1]}

Researchers typically construct county panels across multiple elections (2016, 2020, 2024) to track changes in turnout and vote share over time, separating one-off shocks from sustained trends. This project applies the same

approach—using 2020, 2024, and special elections—to define “shift away from Democrats” as a decline in Democratic turnout, a decline in Democratic vote share, or both.^{[2][1]}

II. Mobilization vs. Persuasion in Electoral Losses

A central question in electoral research is whether party losses stem primarily from mobilization failures (turnout collapse among core supporters) or persuasion losses (voters switching to the other party). Mobilization research emphasizes that campaigns can affect turnout through get-out-the-vote efforts and targeted outreach; field experiments show these effects are often small on average but can be large in competitive contexts and for specific subgroups.^{[3][4]}

Persuasion research, by contrast, focuses on how voters change preferences in response to campaigns, elite rhetoric, media framing, and salient issues like crime, immigration, abortion, and the economy. A decline in vote share alongside stable base turnout points to persuasion loss, while collapsing turnout alongside stable vote share points to mobilization failure.^[4]

Recent studies argue both mechanisms operate simultaneously but their relative importance varies by context: in highly polarized environments mobilization often dominates, since most voters are already “sorted” by party and elections turn on turnout differences, whereas persuasion plays a larger role among swing voters in less polarized contexts.^[4]

For Democratic losses specifically, some analyses suggest turnout collapse among younger, minority, and urban voters drove losses in some regions, while persuasion losses among white working-class and suburban voters mattered more in others—variation a statewide aggregate would obscure.^{[1][4]}

For this project's core question—whether Democratic losses in Pennsylvania reflect turnout collapse, persuasion loss, or both—the literature supports constructing two county-level outcomes, turnout change and vote-share change, to empirically distinguish the two mechanisms: large turnout declines with stable vote share imply mobilization failure, while stable turnout with large vote-share declines implies persuasion loss.^{[3][4]}

III. County Demographics and Political Behavior in Pennsylvania

Pennsylvania is a classic swing state mixing urban, suburban, exurban, and rural counties. Rust-Belt literature emphasizes that county-level demographics—especially education, race, and rural/urban status—are strongly associated with partisan shifts: urban counties like Philadelphia and Allegheny remain strongly Democratic, while many rural and exurban counties have shifted Republican in recent cycles.^{[5][2]}

Turnout also varies systematically: urban counties often have higher absolute turnout but low-income and minority neighborhoods within them sometimes lag more affluent areas, while some rural counties combine consistently high turnout with significant partisan realignment.^{[1][5][2]}

Economic conditions matter too: counties with declining manufacturing employment, opioid crises, or population loss show distinct voting patterns from those with growing service economies, factors typically proxied in quantitative models via median income, education, racial composition, and employment structure.^{[5][1]}

Prior research highlights swing counties in the Philadelphia and Pittsburgh suburbs and central Pennsylvania, which often decide statewide elections and are particularly sensitive to issue salience and campaign messaging—making them well suited to a project linking county-level outcomes to text-based measures of voter attitudes.^{[2][5]}

This literature justifies including county-level demographic controls (education, race, urban/rural status, income) when modeling Democratic shift, and supports focusing on counties that deviate from the state average, since these are likely most informative for understanding mobilization vs. persuasion dynamics.^{[5][1][2]}

IV. Text as Explanation in Political Analysis

A growing body of research uses open-ended survey responses and other textual data to explain political behavior. Such responses let respondents express concerns in their own words, capturing issue salience, emotion, and perceived party performance that closed-ended questions may miss, though analyzing them requires specialized methods.^{[6][7][8]}

Dictionary-based approaches assign words to pre-defined categories (e.g., crime, immigration, abortion, economy) and compute issue emphasis and sentiment scores. These methods are transparent and well-suited to theory-driven research with a clear set of issues of interest—for example, measuring how often respondents mention crime and whether those mentions are positive or negative toward Democrats.^[7]

Topic-modeling approaches (e.g., structural topic models) discover latent themes without pre-defining categories, useful for exploratory analysis and identifying unexpected framing patterns. Recent work applies topic models to open-ended survey items to estimate topic prevalence at the respondent or area level and link topics to outcomes like voting behavior.^[8]

Studies using open-ended responses have linked issue emphasis and sentiment to electoral outcomes: respondents who mention crime frequently with negative sentiment toward a party are more likely to vote against it, with similar patterns for immigration, abortion, and the economy. Aggregated to the county level, these text-derived measures can predict turnout and vote-share changes.^{[7][8]}

In Pennsylvania and similar states, text analysis of open-ended responses can help explain why counties shift away from a party. A county with strong crime emphasis, negative sentiment toward Democrats, and a large vote-share decline is consistent with persuasion loss driven by crime salience; a county with low issue emphasis but a large turnout collapse instead suggests mobilization failure.^{[9][3][4][7][2]}

This project's plan to use an issue-themed dictionary for sentiment on crime, immigration, abortion, and the economy aligns directly with this literature. Regressing county-level Democratic change on demographic and text-derived variables lets us test whether certain issues associate with larger shifts away from Democrats, and whether those shifts fit persuasion or mobilization mechanisms.^{[8][4][7]}

V. Synthesis and Implications for the Project

The literature collectively supports a two-part empirical strategy:

1. **Define and measure "shift away from Democrats"** using county-level turnout and vote-share changes between 2020, 2024, and special elections,. This allows us to distinguish mobilization failure (turnout collapse) from persuasion loss (vote-share decline).^{[3][4][1][2]}
2. **Link county-level shifts to text-derived measures** from open-ended survey responses, using dictionary-based issue emphasis and sentiment on key issues,. This connects voter attitudes and issue salience to electoral outcomes.^{[6][7][8]}

Pennsylvania's heterogeneous county demographics and strong urban–rural divide make it an ideal setting for this analysis. The combination of publicly available election data and open-ended survey text data (e.g., from the Roper Center U.S. State Collection) enables a full start-to-finish data science project that demonstrates both statistical modeling and text-analysis skills.^{[12][13][9][11][2][5]}

VI. Dataset Descriptions

This project examines which Pennsylvania counties shifted away from Democrats in the 2024 presidential election and relevant special elections, and whether those shifts were driven by turnout collapse, persuasion loss, or both. The analysis requires integrating election data (vote totals, turnout, registration), demographic data (Census/ACS), and text data (open-ended survey responses or political discourse). This report describes each required dataset, its background, and all key variables, including their measurement types (categorical, quantitative, ordinal, character).

Dataset 1: Pennsylvania County Election Results and Voter Registration

Source: Pennsylvania Department of State – Voting and Election Statistics (official election returns)

URL: <https://www.pa.gov/agencies/dos/resources/voting-and-elections-resources/voting-and-election-statistics>

Background

The Pennsylvania Department of State (DOS) maintains the official election records for the Commonwealth. The Voting and Election Statistics portal provides public access to summary data for mail ballots, vote history, voter registration by county and party, and election returns. These worksheets contain total registered voters by county, broken down by political party (Democratic, Republican, No Affiliation, and Other), as well as vote totals for major elections.

For this project, the key DOS datasets are:

1. **County-level election returns** for 2020 presidential, 2024 presidential, and selected special elections.
2. **Voter registration worksheets** by county, showing registered voters by party affiliation.

These data are essential for measuring Democratic vote share, turnout rates, and changes over time at the county level. The DOS data are official, publicly available, and updated after each election cycle.

Unit of Analysis

The primary unit of analysis is **county–election**: one row per Pennsylvania county for each election year (e.g., Allegheny County 2020, Allegheny County 2024, etc.). Pennsylvania has 67 counties.

Variables from DOS Election and Registration Data

The following variables are derived from the DOS election returns and registration worksheets:

Variable Name	Description	Type
county	Name of the Pennsylvania county (e.g., “Allegheny”, “Philadelphia”)	Character
county_fips	County FIPS code (unique identifier for each county)	Character / Categorical
election_year	Year of the election (e.g., 2020, 2024)	Quantitative (discrete)
dem_votes	Total votes cast for Democratic candidate(s) in this county and election	Quantitative (discrete, count)
gop_votes	Total votes cast for Republican candidate(s) in this county and election	Quantitative (discrete, count)
other_votes	Total votes for third-party or other candidates	Quantitative (discrete, count)
total_votes	Total votes cast in this county and election (sum of all party votes)	Quantitative (discrete, count)
registered_dem	Number of registered Democratic voters in the county	Quantitative (discrete, count)
registered_gop	Number of registered Republican voters in the county	Quantitative (discrete, count)
registered_no_affiliation	Number of registered voters with no party affiliation	Quantitative (discrete, count)
registered_other	Number of registered voters in “Other” categories	Quantitative (discrete, count)
registered_total	Total registered voters in the county	Quantitative (discrete, count)

From these raw variables, I will attempt to construct analytical variables:

Variable Name	Formula / Description	Type
dem_vote_share	$\text{dem_votes} / \text{total_votes}$ (Democratic share of total votes, 0–1)	Quantitative (continuous)
gop_vote_share	$\text{gop_votes} / \text{total_votes}$ (Republican vote share, 0–1)	Quantitative (continuous)
turnout_rate	$\text{total_votes} / \text{registered_total}$ (proportion of registered voters who voted)	Quantitative (continuous, 0–1)
dem_turnout_share	Conceptual: $\text{dem_votes} / (\text{registered_dem} * \text{eligibility_factor})$	Quantitative (continuous)
dem_vote_share_2020	Democratic vote share in 2020 presidential election	Quantitative (continuous)
dem_vote_share_2024	Democratic vote share in 2024 presidential election	Quantitative (continuous)
dem_voteshare_change	$\text{dem_vote_share_2024} - \text{dem_vote_share_2020}$	Quantitative (continuous, can be negative)
turnout_change	$\text{turnout_rate_2024} - \text{turnout_rate_2020}$	Quantitative (continuous, can be negative)
shift_away_from_dem	Numeric measure of Democratic decline (e.g., positive if dem share dropped)	Quantitative (continuous)
shift_category	Categorical label: “large decline”, “moderate decline”, “stable”, “gain”	Categorical (ordinal)

The key outcome variable is **shift_away_from_dem**, which quantifies the Democratic decline in each county. Counties are then ranked or categorized by how much they declined relative to the state average or by absolute magnitude of drop.

Dataset 2: National Institute on Minority Health and Health Disparities Data

Source: NIMHD HDPulse Data Portal and related health disparity indicators

URL: <https://hdpulse.nimhd.nih.gov/index.html>

Background

The National Institute on Minority Health and Health Disparities provides a comprehensive source of public health and health equity indicators for U.S. populations and communities. The HDPulse portal offers data on social, economic, environmental, and health-related factors, including measures related to access to care, chronic disease burden, neighborhood conditions, and population-level disparities by race, ethnicity, and geography. For this project, the county-level or place-based indicators available through the portal are preferred because they provide more localized measures for comparing Pennsylvania counties.

These health disparity variables are used as control variables and explanatory factors in regression analyses of Democratic vote-share change. They help determine whether shifts are associated with community health conditions, access to resources, social disadvantage, or place-based inequities rather than only traditional demographic composition. Dataset 2: U.S. Census Bureau Demographic and Socioeconomic Data

Unit of Analysis

The unit of analysis is **county** (one row per Pennsylvania county). These data are merged with the election dataset using `county_fips` or `county_name`.

Variables considered from the NIH HDPulse portal data

Variable Name	Description	Type
<code>pct_ba</code>	Percent with bachelor's degree	Quantitative, continuous
<code>pct_bafemale</code>	Percent of females with bachelor's degree	Quantitative, continuous
<code>pct_bamale</code>	Percent of males with bachelor's degree	Quantitative, continuous
<code>pct_bablack</code>	Percent of Black population with bachelor's degree	Quantitative, continuous
<code>pct_bawhite</code>	Percent of White population with bachelor's degree	Quantitative, continuous
<code>income</code>	Median household income	Quantitative, continuous
<code>pct_popbelow18</code>	Percent of population below age 18	Quantitative, continuous
<code>pct_pop18to39</code>	Percent of population age 18 to 39	Quantitative, continuous
<code>pct_pop40up</code>	Percent of population age 40 and older	Quantitative, continuous
<code>pct_black</code>	Percent Black	Quantitative, continuous
<code>Urban_rural</code>	Designation of county as either urban or rural	Categorical
<code>pct_foreignborn</code>	Percent foreign born	Quantitative, continuous
<code>pct_white</code>	Percent White	Quantitative, continuous
<code>pct_poverty</code>	Percent below poverty line	Quantitative, continuous
<code>pct_unemployed</code>	Percent unemployed	Quantitative, continuous

These variables are primarily **quantitative** (continuous percentages or counts). They serve as explanatory variables in regressions predicting `dem_voteshare_change` or `shift_away_from_dem`.

Dataset 3: County-Level News Sentiment (GDELT Project, Aggregated by Media Market)

Primary Source: GDELT Project (Global Database of Events, Language, and Tone)

URL: <https://www.gdeltproject.org/>

Background

The original plan for this dataset was open-ended survey text from Pennsylvania respondents — specifically Pew Research Center’s American Trends Panel (ATP) and the American National Election Studies (ANES) — filtered to Pennsylvania and analyzed for issue sentiment. In practice, neither source supports this: ATP’s public microdata carries no state variable at all (only Census region, division, and a metro/non-metro flag), so it cannot be filtered to Pennsylvania, and ANES does include a state identifier but redacts county-level fields, making county-resolved analysis possible only through a restricted-access ICPSR agreement covering roughly 100–250 Pennsylvania respondents. Twitter/X and other social-media sources were also evaluated and ruled out for related geographic-resolution reasons. See Selecting a Text and Sentiment Data Source later in this report for the full comparison of alternatives considered.

The text/sentiment data actually used comes from the **GDELT Project**’s public news archive. For four salient 2024 campaign issues — crime, immigration, abortion, and the economy — GDELT tone scores were pulled for January–November 2024 and aggregated to Pennsylvania’s 11 Nielsen media markets (designated market areas, or DMAs), since most individual Pennsylvania counties do not generate enough dedicated local news coverage to support a reliable county-level tone measure on their own. Each county is then assigned its media market’s tone scores via a published county-to-DMA crosswalk, so every county sharing a market receives the same four tone values. These four tone measures serve directly as the text/sentiment explanatory variables in the Question 3 extended ridge regression model.

Variables from County-Level News Sentiment (GDELT)

Variable Name	Description	Type
TVDMA	Nielsen designated media market area (DMA) name assigned to the county via the county-to-DMA crosswalk	Character / Categorical
crime_tone	Average GDELT news tone score for crime coverage in the county’s media market, Jan.–Nov. 2024	Quantitative (continuous)
immigration_tone	Average GDELT news tone score for immigration coverage in the county’s media market	Quantitative (continuous)
abortion_tone	Average GDELT news tone score for abortion coverage in the county’s media market	Quantitative (continuous)
economy_tone	Average GDELT news tone score for economy coverage in the county’s media market	Quantitative (continuous)
text_data_imputed	Indicator: 1 if a market/issue tone value was too sparse and was replaced with the cross-market median, 0 otherwise	Categorical (binary)

Dataset 4: Geographic Boundary Files (Shapefiles)

Source: U.S. Census Bureau TIGER/Line Shapefiles; Redistricting Data Hub

URL: <https://redistrictingdatahub.org/state/pennsylvania/>

Background

To create geographic visualizations (maps) shading counties by Democratic shift, turnout change, or dominant issue, we need county boundary shapefiles. The U.S. Census Bureau's **TIGER/Line Shapefiles** provide official geographic boundaries for U.S. counties, including FIPS codes that can be joined to election and demographic data. The Redistricting Data Hub also provides ready-to-use county-level boundary files and CVAP data for Pennsylvania.

Unit of Analysis

The unit of analysis is **county** (one polygon per Pennsylvania county).

Variables from Shapefiles

Variable Name	Description	Type
county_fips	County FIPS code (unique identifier)	Character / Categorical
county_name	County name	Character
geometry	Spatial geometry (polygon boundaries)	Spatial / Character (WKT)
state_fips	State FIPS code for Pennsylvania (42)	Character / Categorical

These variables are primarily **character** (FIPS codes, names) and **spatial** (geometry). The shapefile is used in GIS software or R/Python mapping libraries to create choropleth maps shading counties by `shift_away_from_dem`, `dem_vote_share_change`, or an issue-tone measure.

Integration of Datasets

The final analytical dataset combines these four sources:

1. **Election data** (DOS) provides the core outcome variables (`dem_vote_share_change`, `shift_away_from_dem`, `turnout_change`).
2. **Demographic data** (Census/ACS) provides explanatory variables (`pct_white`, `pct_college_educated`, `median_household_income`, etc.).
3. **Text data** (GDELT news archive, aggregated to Pennsylvania's 11 media markets) provides issue-tone variables (`crime_tone`, `immigration_tone`, `abortion_tone`, `economy_tone`).
4. **Shapefiles** enable geographic visualization.

The master dataset has one row per **county**, with columns for election outcomes, demographics, and text-derived sentiment measures. A regression model might take the form:

$$\text{dem_vote_share_change}_i = \beta_0 + \beta_1 \text{demo}_i + \beta_2 \text{text_sentiment}_i + \epsilon_i$$

where i indexes counties.

VI. ANALYSIS

Research Questions and Related Hypotheses

This project investigates county-level shifts away from the Democratic Party in Pennsylvania during the 2024 presidential election and related special elections, asking which counties experienced the largest declines, whether those losses were driven primarily by turnout collapse or by voters switching parties, and the extent to which county socioeconomic characteristics and issue-focused political discourse explain observed variation. Specifically, the study addresses three research questions: (1) Which Pennsylvania counties experienced the largest shifts away from the Democratic Party in the 2024 presidential election and related special elections? (2) Were Democratic losses in Pennsylvania counties driven primarily by turnout decline, vote-share decline (persuasion), or a combination of both? and (3) To what extent do county demographics and issue-based text measures explain variation in county-level Democratic decline across Pennsylvania? I hypothesize that (H1) Democratic decline will be heterogeneously distributed with larger losses concentrated in suburban, exurban, and select rural counties; (H2) both mobilization failure (turnout decline) and persuasion (vote-share decline) will contribute to Democratic losses, with the dominant mechanism varying by county; and (H3) counties with lower educational attainment, lower median incomes, higher shares of White voters, and stronger negative sentiment toward Democrats on issues such as crime, immigration, abortion, and the economy will exhibit larger shifts away from Democrats.

Results: Research Question 1

Which Pennsylvania counties experienced the largest shifts away from the Democratic Party in the 2024 presidential election and related special elections, and by how much?

Overview

Across all 67 Pennsylvania counties, the shift in Democratic vote share from 2020 to 2024 was broadly negative but far from uniform. Forty-six counties (69%) recorded a decline in Democratic vote share, one county (Mifflin) was effectively unchanged (0.000), and twenty counties (30%) recorded a modest gain. The magnitude of decline ranged from a fraction of a percentage point up to 3.4 percentage points, with the losses heavily concentrated in a relatively small number of counties rather than spread evenly across the Commonwealth (Figures 1 and 2).

Magnitude and Ranking of the Shift

Figure 1 ranks all 67 counties by their change in Democratic vote share. The ten largest declines and the ten largest gains are summarized in Tables 1 and 2 below.

Table 1. Ten largest Democratic vote-share declines, 2020–2024

Rank	County	Δ Dem. vote share
1	Monroe	−0.034
2	Lackawanna	−0.027

Rank	County	Δ Dem. vote share
3	Philadelphia	−0.026
4	Lehigh	−0.025
5	Pike	−0.024
6	Luzerne	−0.023
7	Bucks	−0.022
8	Fayette	−0.021
9	Clinton	−0.021
10	Lawrence / Mercer (tied)	−0.019

Table 2. Ten largest Democratic vote-share gains, 2020–2024

Rank	County	Δ Dem. vote share
1	Cumberland	+0.007
1	Perry	+0.007
1	Union	+0.007
4	Sullivan	+0.006
4	Warren	+0.006
6	Jefferson	+0.005
6	Adams	+0.005
6	Snyder	+0.005
9	Montour	+0.004
9	Butler	+0.004

Notably, even the largest gains (0.7 points) were an order of magnitude smaller than the largest losses (3.4 points), underscoring that the dominant story statewide was decline, not a symmetric two-directional shift.

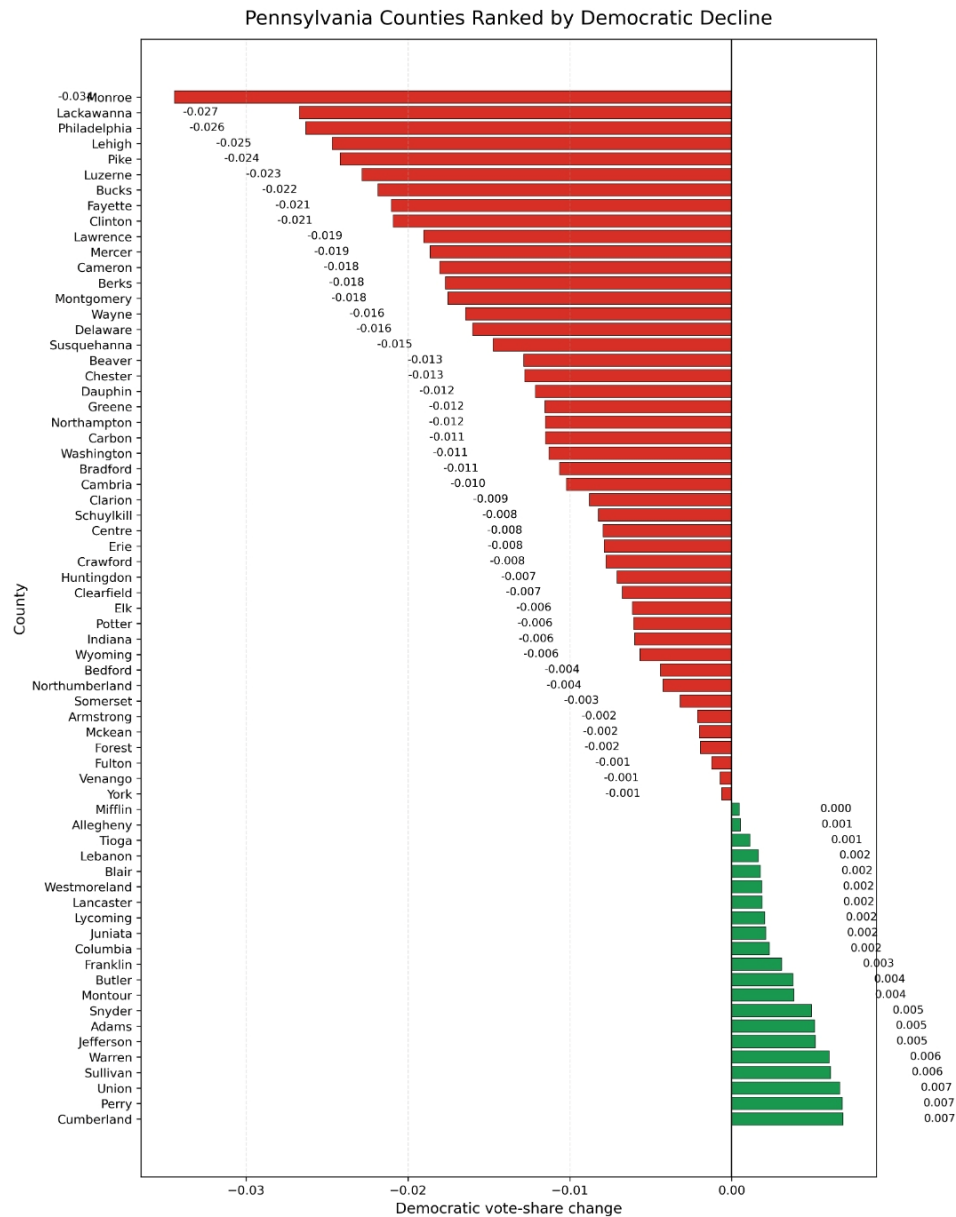


Figure 1. Pennsylvania counties ranked by change in Democratic vote share, 2020–2024.

Geographic Pattern

The statewide choropleth map in Figure 2 confirms and sharpens the picture from the ranking: Democratic decline was not scattered randomly across Pennsylvania, but formed three distinct, geographically contiguous clusters, with a fourth contiguous cluster of gains in the south-central part of the state.

Pennsylvania County Democratic Shift 2020 - 2024

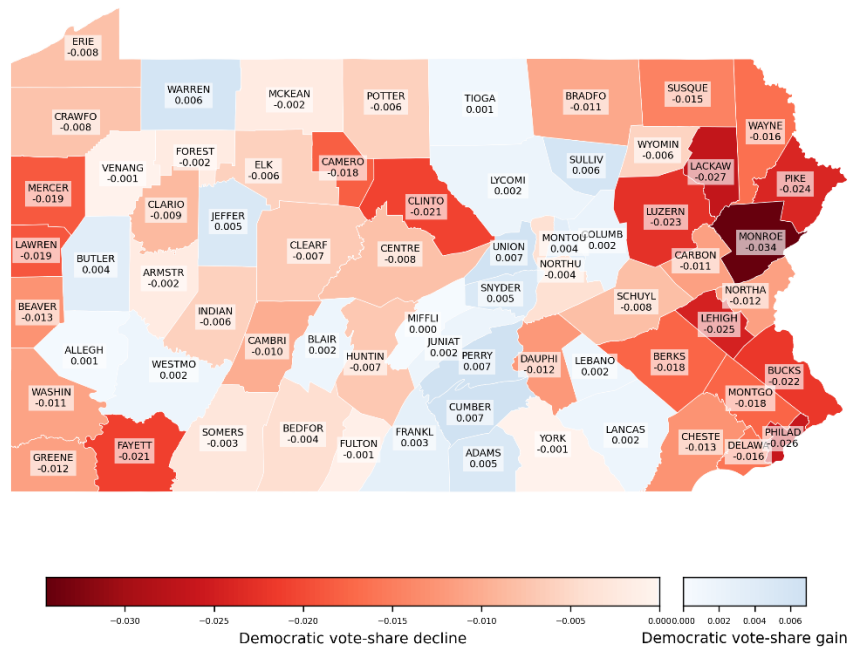


Figure 2. Statewide choropleth of Democratic vote-share change by county, 2020–2024.

- Northeastern PA (Poconos/Scranton corridor): Monroe (−0.034), Lackawanna (−0.027), Pike (−0.024), Luzerne (−0.023), Wayne (−0.016), and Susquehanna (−0.015) form a contiguous band of the darkest red on the map, marking the single most intense zone of Democratic erosion in the state.
- Southeastern Philadelphia corridor: Philadelphia (−0.026), Lehigh (−0.025), Bucks (−0.022), Montgomery (−0.018), Berks (−0.018), Delaware (−0.016), Chester (−0.013), and Northampton (−0.012) form a second contiguous cluster spanning the city and its full suburban ring.
- Southwestern industrial belt: Fayette (−0.021), Lawrence (−0.019), Mercer (−0.019), Beaver (−0.013), Washington (−0.011), Greene (−0.012), and Cambria (−0.010) form a third contiguous cluster along the western border, consistent with continued erosion in Pennsylvania's former industrial counties.
- South-central "gains" cluster: Cumberland (+0.007), Perry (+0.007), Union (+0.007), Adams (+0.005), and Snyder (+0.005) form the only contiguous block of counties with positive Democratic movement on the map, centered on the Harrisburg / Susquehanna Valley area.
- A handful of gain counties are geographically isolated rather than clustered — Warren (+0.006) and Sullivan (+0.006) sit in the north-central forest region and Butler (+0.004) sits in the southwest, each surrounded by counties that declined. This suggests county-specific dynamics rather than a broader regional trend in those cases, a distinction worth revisiting under Question 3.

Vote-Share Shift Versus Turnout

Figures 3 through 5 provide descriptive context for how this decline relates to turnout, which will be examined formally under Question 2. Each scatterplot below applies a small amount of random jitter to each point's plotted position — never to the underlying data — so that county labels can be read without excessive overlap. In the percentage-change

view (Figure 3), Republican vote share rose in nearly every county (red points mostly above 0) while Democratic vote share fell (blue points mostly below 0), and this separation between the two series holds fairly consistently across the full range of turnout change (roughly -5% to $+12\%$), reinforcing that the vote-share shift was not simply a byproduct of which counties turned out more or less. With county labels now visible, several individual counties stand out. Monroe combines the state's single largest Democratic vote-share decline (-3.4 points) with its single largest Republican vote-share gain ($+3.7$ points), placing it furthest from the origin of any county. Cameron sits in visual isolation on the far left, pairing the state's steepest turnout decline (-4.6 points) with a vote-share shift that still moved against Democrats, showing that lower turnout alone does not explain the direction of the shift. At the opposite extreme, Warren, Jefferson, and Montour cluster together on the far right with the state's largest turnout increases ($+9.1$ to $+11.4$ points) but only modest vote-share movement, indicating that the counties with the biggest turnout gains were not where the partisan shift was largest.

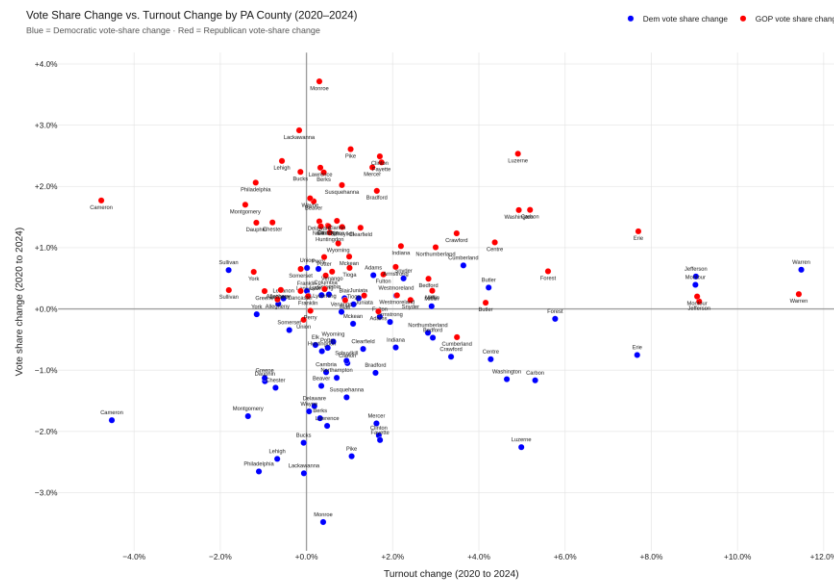


Figure 3. Democratic and Republican vote-share change versus turnout change by county, 2020–2024 (jittered; all 67 counties labeled).

The raw vote-count view tells a different story once population size enters the picture. Plotting all 67 counties on the same axis (Figure 4) shows that Philadelphia's raw Democratic vote count fell by 35,219 votes — more than five times the next-largest Democratic decline (Bucks, $-6,281$) — which compresses every other county into a narrow band near zero and makes county-to-county variation among the remaining 66 counties difficult to see. Montgomery County stands out even at this compressed scale: despite a comparatively modest vote-share shift, it added 12,851 raw Republican votes, the single largest raw Republican gain of any county statewide, illustrating how a large, dense county can generate substantial raw vote swings without necessarily posting the state's largest percentage shift.

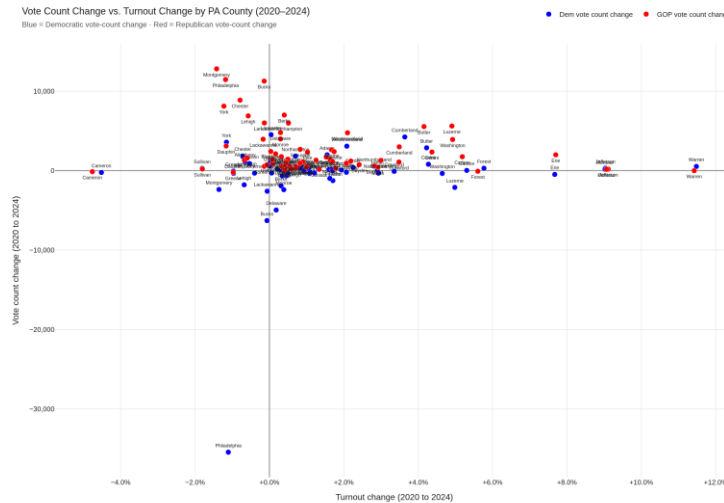


Figure 4. Democratic and Republican raw vote-count change versus turnout change by county, 2020–2024 (all 67 counties, jittered and labeled).

Removing Philadelphia’s outsized swing (Figure 5) rescales the chart and reveals the substructure among the remaining 66 counties that Figure 4 obscured. Suburban Philadelphia counties dominate the Republican side of the raw vote-count ledger: Montgomery (+12,851), Bucks (+11,355), Chester (+8,734), and York (+8,151) each added thousands of Republican votes even where turnout change was near zero, while Democratic totals in those same counties moved far less — Bucks fell by 6,281 votes, the largest Democratic raw decline among the remaining counties, and most others stayed within roughly $\pm 5,000$. This pattern indicates that a meaningful portion of the statewide shift reflects existing or newly cast votes moving toward the Republican column rather than a pure turnout effect, foreshadowing the mobilization-versus-persuasion analysis in Question 2.

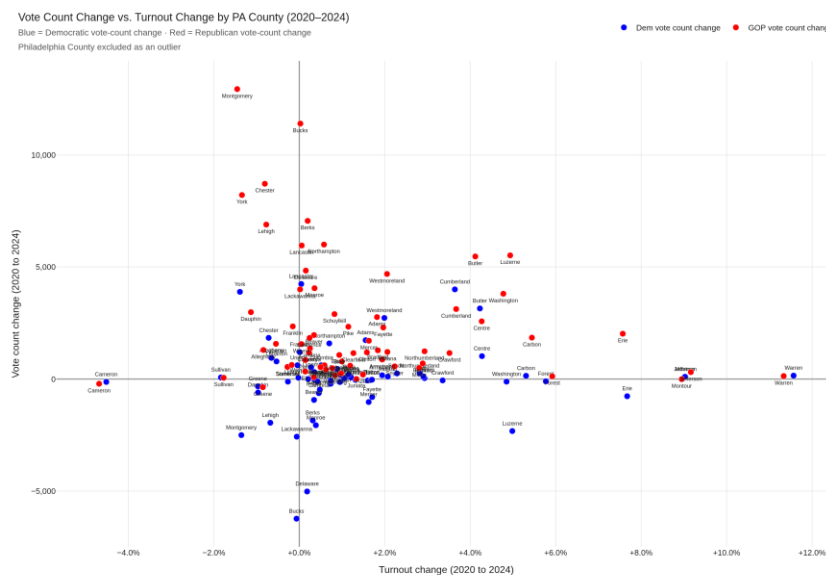


Figure 5. Democratic and Republican raw vote-count change versus turnout change by county, 2020–2024 (Philadelphia excluded as an outlier; jittered and labeled).

Summary

The data support a clear answer to Question 1: the shift away from the Democratic Party was not uniform across Pennsylvania, but formed three contiguous regional clusters of decline — the northeastern Poconos/Scranton corridor (led by Monroe, -0.034), the greater Philadelphia region (Philadelphia, Bucks, Lehigh), and the southwestern industrial belt (Fayette, Lawrence, Luzerne) — while a single contiguous block of already-Republican south-central counties (Cumberland, Perry, Union) was the only area to see the Democratic share tick upward. The magnitude of decline in the hardest-hit counties (2.5–3.4 percentage points) was several times larger than the gains recorded anywhere else in the state.

Results: Research Question 2

Were Democratic losses in Pennsylvania counties driven primarily by turnout decline, vote-share decline (persuasion), or a combination of both?

Following the operational definitions established above, counties with large Democratic turnout declines but stable vote share suggest mobilization failure; counties with stable turnout but large Democratic vote-share declines suggest persuasion loss^[4]; counties with large declines in both dimensions suggest that the two mechanisms are compounding one another^[3]. This analysis operationalizes those definitions using the same 67-county Pennsylvania election dataset built for Question 1 and tests H2 directly.

Method

Two county-level outcome variables carry the analysis: `turnout_change` (share of registered/eligible voters casting a ballot, 2020 to 2024) and `demvoteshare_change` (Democratic share of the two-party vote, 2020 to 2024), drawn from the same 67-county Pennsylvania election dataset used for Question 1.

A four-category county typology was built by cross-classifying each county's position on both variables against their own quartile distributions: values at or below the 25th percentile are labeled “large decline,” values above the 75th percentile are labeled “increase/gain,” and the middle 50% is labeled “stable.” A county is classified Mobilization-Driven Decline when turnout is in large decline but vote share is stable; Persuasion-Driven Decline when turnout is stable (or increasing) but vote share is in large decline; Mixed when both are in large decline; Resilient / Gain when vote share increased regardless of turnout; and Minimal Change / Ambiguous otherwise.

To complement the categorical typology with a continuous test of H2, a Pearson correlation and an ordinary least squares (OLS) regression of `demvoteshare_change` on `turnout_change` were estimated, with a second model adding a rural/urban indicator as a covariate. Regression residuals were then examined directly: counties whose actual vote-share decline is far larger than turnout alone would predict are the clearest cases of persuasion loss operating independently of mobilization.

Overview

Figure 6 maps the geographic distribution of `turnout_change` across all 67 counties. Turnout increases (blue) cluster in a mix of rural northern-tier and small urban counties — Warren (+11.4 points), Jefferson (+9.1), Montour (+9.1), and Erie (+7.5) posted the largest gains — while turnout declines (orange) are more scattered, led by Cameron (-4.6 points), Sullivan (-1.7), and Greene (-1.0). This spatial pattern is the raw mobilization signal that the typology below cross-references against Democratic vote-share change.

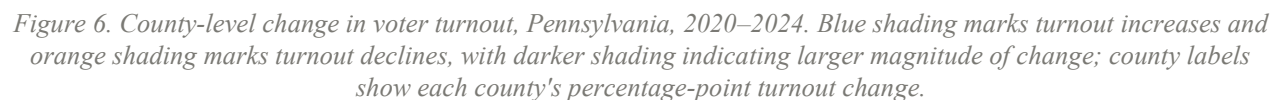


Table 3 summarizes the number of counties in each typology category, along with each group's average turnout change and average Democratic vote-share change.

Typology	N counties	Mean turnout change	Mean Dem. vote-share change
Persuasion-Driven Decline	11	+1.2%	-2.1%
Mobilization-Driven Decline	7	-0.8%	-0.5%
Mixed (Mobilization + Persuasion)	6	-1.3%	-2.3%
Resilient / Gain	17	+2.6%	+0.4%
Minimal Change / Ambiguous	26	+2.2%	-0.6%

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Statistical Relationship Between Turnout and Vote-Share Change

A Pearson correlation between turnout_change and demvoteshare_change across all 67 counties is positive and statistically significant ($r = 0.274$, $p = 0.025$), indicating that counties with larger turnout increases tended to see somewhat smaller Democratic vote-share declines (or larger gains). However, the corresponding R^2 of 0.075 means turnout change explains only about 7.5% of the variance in vote-share change — the remaining 92.5% is driven by factors other than turnout, which is itself evidence that persuasion-type mechanisms (or omitted variables such as demographics and issue salience, to be tested under Question 3) account for the majority of the shift.

Adding a rural/urban indicator as a covariate raises the model's explanatory power modestly, to an R^2 of 0.120. Urban counties show an additional average vote-share decline of about 0.5 percentage points relative to rural counties after accounting for turnout (marginally significant, $p = 0.075$), suggesting urban counties experienced somewhat more persuasion-type loss than their turnout alone would predict.

Table 4. OLS regression of Democratic vote-share change on turnout change, with and without a rural/urban covariate.

Model	turnout_change coef.	p-value	R ²
demvoteshare_change ~ turnout_change	0.103	0.025	0.075
+ rural/urban covariate	0.086	0.060	0.120

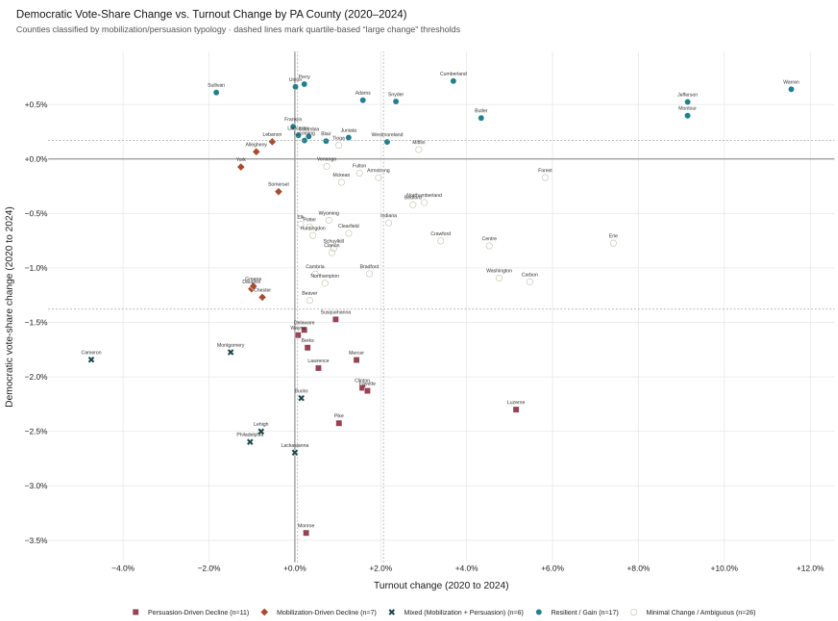


Figure 7. Democratic vote-share change versus turnout change by county, 2020–2024, classified by mobilization/persuasion typology (jittered; all 67 counties labeled; dashed lines mark quartile-based thresholds).

Outlier Counties: Where Persuasion Exceeds What Turnout Predicts

Regression residuals identify the counties where the simple turnout-based model most under- or over-predicts the actual vote-share change. The six counties with the largest negative residuals — Monroe, Luzerne, Lackawanna, Pike, Philadelphia, and Lehigh — lost far more Democratic vote share than their turnout change alone would predict, making

them the clearest cases of persuasion loss operating on top of (or independent from) any turnout effect. Monroe is the most extreme case: turnout there barely changed (+0.4 points), yet Democratic vote share fell 3.4 points, the largest decline statewide.

At the other extreme, six counties — Sullivan, Union, Perry, Adams, Franklin, and Cumberland — held up noticeably better than their turnout change would suggest, several of them posting vote-share gains despite flat or even declining turnout. These counties are the strongest evidence against a simple mobilization-only story: turnout alone does not explain why some counties resisted the statewide Democratic decline while others facing similar turnout conditions did not.

Discussion

The evidence supports H2's claim that both mobilization and persuasion mechanisms contributed to Pennsylvania's Democratic decline, with the dominant mechanism varying by county. But the two mechanisms are not equally represented: among counties with a clear, large decline, persuasion-type patterns (11 counties) are more common than pure mobilization failure (7 counties), and the turnout-only regression model explains a small share of the statewide variance in vote-share change ($R^2 = 0.075\text{--}0.120$). Six counties, including Philadelphia, Lackawanna, Lehigh, Bucks, Montgomery, and Cameron, show both mechanisms compounding simultaneously — a meaningfully large group given that only 24 of 67 counties displayed a clear single-mechanism decline at all.

These findings should be read as descriptive and correlational rather than causal: the quartile-based thresholds used to build the typology are one reasonable and replicable choice among several, and a sensitivity check using alternative cut-points (e.g., median splits or one-standard-deviation bands) would be a useful robustness extension. Turnout and vote-share change are also measured over the same four-year window, so the regression cannot rule out reverse influence (e.g., a candidate's declining local support also suppressing turnout among discouraged supporters).

These results motivate Question 3: if persuasion-type decline is the more common pattern among Pennsylvania's most affected counties, then county demographics and issue-based text measures — particularly sentiment on crime, immigration, abortion, and the economy — become the natural next explanatory layer for why voters in counties like Monroe, Luzerne, and Philadelphia moved away from the Democratic Party even where turnout stayed flat or rose. Data are drawn from Pennsylvania county-level election and turnout records for the 2020 and 2024 general elections^[12].

Results: Research Question 3

To what extent do county-level demographic characteristics explain variation in Democratic vote-share change across Pennsylvania's 67 counties?

Question 2 found that persuasion-type decline — voters remaining engaged but shifting their vote choice — was the more common pattern among counties with the largest Democratic losses, and that a simple turnout-only model left the large majority of that variance unexplained ($R^2 = 0.075\text{--}0.120$). The governing hypothesis (H3) states that county demographic composition — educational attainment, income, race, poverty, unemployment, and foreign-born population share — accounts for a meaningfully larger share of that persuasion-driven variance than turnout alone.

This section presents the demographics-only stage of the analysis. A later subsection below extends this model with issue-based news sentiment measures — coverage tone on crime, immigration, abortion, and the economy, aggregated by media market — once the process of selecting and validating a suitable text data source is described; the demographic

model presented here stands on its own as an estimate of how much of Pennsylvania’s county-level Democratic decline is explained by demographic composition alone.

Method

Six county-level demographic variables drawn from the American Community Survey^[14] — percent with a bachelor’s degree or higher, median household income, percent white, poverty rate, unemployment rate, and percent foreign-born — were merged with the same 67-county Pennsylvania election dataset used for Questions 1 and 2, using the outcome variable Democratic share of the two-party vote, 2020 to 2024.

Because income, education, and poverty are structurally correlated at the county level, a variance inflation factor (VIF) check was run on the raw predictors first as a diagnostic. VIF values well above the conventional 5–10 threshold confirmed meaningful multicollinearity, motivating Ridge regression^[15] over ordinary least squares: Ridge shrinks correlated predictors together in a stable way, whereas OLS coefficients would be unstable and sensitive to small data changes at this sample size ($n = 67$).

Predictors were standardized (z-scored) prior to fitting, since Ridge’s penalty term is scale-dependent. RidgeCV was then fit with leave-one-out cross-validation (LOOCV) to select the penalty parameter (alpha) from a grid of 100 log-spaced candidates (0.01 to 10,000); leave-one-out was chosen over standard k-fold cross-validation because k-fold estimates are unstable at $n = 67$. Model fit was evaluated two ways: in-sample R^2 (how well the model fits the data it was trained on) and LOOCV R^2 and RMSE (the honest, out-of-sample estimate, obtained by predicting each county from a model that never saw that county during fitting).

Multicollinearity Diagnostic

Table 5 reports the VIF for each raw predictor. All six exceed the conventional multicollinearity threshold, with percent white and median household income showing especially severe overlap with the rest of the predictor set — exactly the situation Ridge’s shrinkage is designed to handle.

Variable	VIF
White population share	176.08
Median household income	175.46
Poverty rate	42.21
Bachelor’s degree+	34.61
Unemployment rate	19.62
Foreign-born share	16.94

Table 5. Variance inflation factors for the six raw demographic predictors. Values above 5–10 indicate meaningful multicollinearity.

Ridge Model

Leave-one-out cross-validation selected a penalty value of $\alpha = 9.326$. Figure 8 shows the six standardized coefficients ranked by magnitude. Unemployment rate and foreign-born population share are the two strongest predictors, each roughly 1.5–3 times larger in magnitude than the next three predictors, and both carry a negative sign: counties with higher unemployment and a higher foreign-born share saw larger Democratic vote-share declines, net of the other five demographic variables.

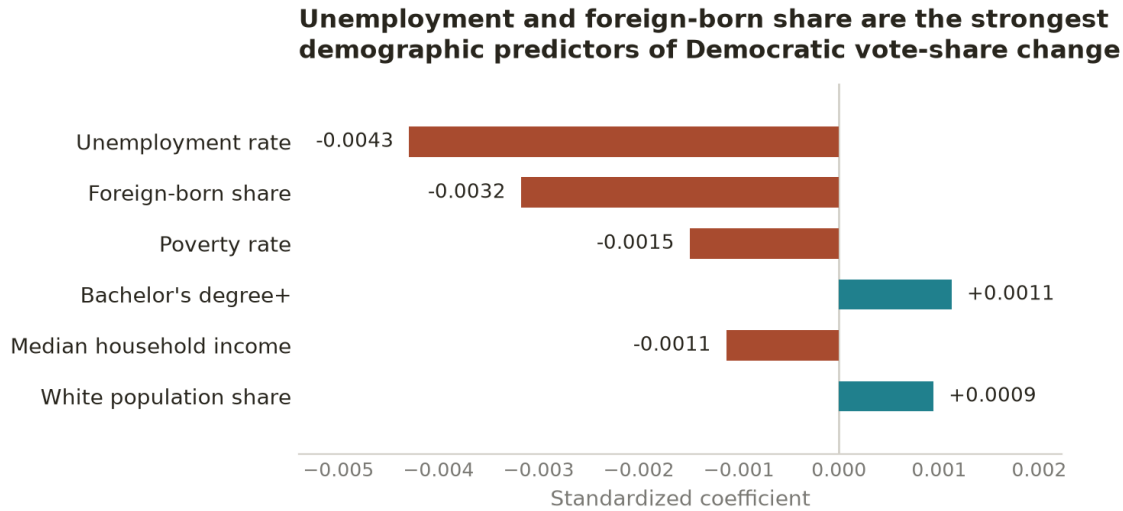


Figure 8. Standardized Ridge regression coefficients, demographics-only model. Bars show direction and relative magnitude; all predictors are z-scored, so magnitudes are directly comparable.

Metric	Value
Selected alpha (lambda)	9.326
In-sample R^2	0.569
LOOCV R^2 (out-of-sample)	0.479
LOOCV RMSE	0.00721

Table 6. Ridge model fit statistics, demographics-only model. LOOCV R^2 is the honest out-of-sample estimate and should be read as the primary result.

Discussion

County demographic composition alone explains a substantial share of Pennsylvania’s county-level Democratic vote-share change: the LOOCV R^2 of 0.479 means demographics account for roughly 48% of the honest, out-of-sample variance — a large jump from the 7.5–12% explained by turnout alone in Question 2, and consistent with H3’s claim that persuasion-type decline is substantially demographic in character.

The two strongest predictors — unemployment rate and foreign-born population share — both point in the same direction: counties with more economic distress and larger immigrant populations saw larger Democratic vote-share declines. Poverty rate and median household income point the same way (negative), while education (bachelor’s degree+) and racial composition (percent white) show smaller, positive coefficients. Notably, the two income-and-poverty-adjacent variables (median household income and poverty rate) do not simply cancel each other out or wash out under Ridge’s shrinkage — both retain meaningful, same-signed coefficients even after accounting for their mutual correlation, suggesting each captures a distinct dimension of economic distress rather than redundant information.

Roughly half of the out-of-sample variance remains unexplained by demographics alone. Closing that gap is the motivation for the text-based sentiment extension presented later in this section: if certain counties’ local press coverage of crime, immigration, abortion, or the economy diverged sharply from what their demographic profile alone would predict, that divergence is exactly the kind of signal a text-based measure could pick up that a purely demographic model cannot.

As with Question 2, these results should be read as descriptive and correlational, not causal. County-level demographic averages are ecological measures — they describe the composition of a county’s population, not the behavior of any individual voter within it — so caution is warranted against inferring individual-level mechanisms from these county-level associations. The severe multicollinearity documented in Table 5 also means that while Ridge produces stable, interpretable coefficients as a set, the precise magnitude of any single coefficient is less certain than an OLS model would (misleadingly) suggest; the ranking and sign of the top predictors is more robust than their exact numeric values.

Selecting a Text and Sentiment Data Source

H3’s original framing called for issue-based sentiment drawn from open-ended survey text. Before building any pipeline, several candidate sources were tested for one specific requirement: the ability to resolve to Pennsylvania county-level (or at least sub-state) geography. A source with strong topical fit but no usable geography was not a viable predictor for this analysis, regardless of its academic pedigree.

Survey Microdata: Ruled Out Early

The most direct reading of the proposal — open-ended responses from an established academic panel such as ANES or the Cooperative Election Study — was ruled out first. Public-use releases of these studies strip county-level geography for respondent privacy; obtaining a version with county identifiers requires a restricted-access agreement, typically through ICPSR, which involves institutional sponsorship and an approval process measured in weeks rather than days. For a project on this timeline, pursuing restricted access before any other option had been tried was not an efficient starting point.

News-Based Alternatives

The first tractable substitute tested was the GDELT Project^[17]’s public news archive: a free, keyless API that can be filtered by exact historical date range and by keyword, with article-level tone scores that aggregate into an issue-level sentiment measure. A feasibility check across a large county (Erie, roughly 270,000 residents) and two small ones (Bedford, roughly 48,000; Forest, roughly 7,000, the state’s smallest) confirmed the API worked as advertised but exposed a coverage problem: Erie returned a usable ~81 articles for the immigration query alone, while Bedford returned only ~9 articles across all four issues combined, and Forest’s ~29 hits were almost entirely unrelated national wire stories caught by loose keyword matching rather than genuine local reporting. With more than 40 of Pennsylvania’s 67 counties in this small/rural category, a county-level version of this measure would be mostly noise. This sparsity problem — and the media-market aggregation strategy adopted to solve it — is described in the Text Data Construction subsection below.

Before committing to that aggregation strategy, a range of other text sources were screened for whether any could deliver genuine county-level (rather than market-level) resolution. None cleared the bar without a cost or delay disproportionate to the likely gain (Table 7).

Source	What it offers	Verdict
Twitter/X	Full-archive historical search, priced per call.	Ruled out — geotagging was largely removed from the default posting flow years ago; the small, self-selecting share of posts that remain geotagged skews younger, more urban, and more politically online than the general electorate, which would likely produce a sparser and more biased sample than the news-based approach.

Source	What it offers	Verdict
Media Cloud	Free academic platform (UMass Amherst/Northeastern) built for local-news research, with a place-tagging engine designed to reduce national-wire noise.	Not pursued — fully account-gated, with no anonymous querying; registration requires a written justification and email confirmation before any query can be tested, adding real delay for an uncertain improvement over the working GDELT measure.
Local-outlet targeting (NELA-Local method)	Restrict queries to each county’s own local paper’s domain (e.g., the Bedford Gazette for Bedford County) rather than broad keyword search, borrowing a mapping technique from an academic local-news dataset.	Promising but labor-intensive — would require identifying and validating roughly 67 outlet domains by hand; the reference dataset itself only covers April 2020–December 2021, so it is a method to borrow, not a ready-made 2024 dataset.
Google Trends	Search-interest-by-region for issue keywords.	Ruled out — geographic granularity is capped at the metro/media-market level, the same resolution the GDELT aggregation already uses, so it offers no improvement.
Reddit	County- and city-specific subreddits.	Ruled out — most small Pennsylvania counties have no dedicated subreddit, and the user base that exists skews young and urban, the same representativeness problem as Twitter/X.
Meta Content Library	Public-content API spanning Facebook, Instagram, and Threads, including Facebook Groups, which could reach genuine local community discussion.	Ruled out for this project — restricted to academic/nonprofit use with an ICPSR-style vetting process taking four to six weeks, plus a setup and monthly fee under at least one access path; realistic only with an active university research affiliation and a longer timeline.

Table 7. Alternative text/sentiment sources evaluated before proceeding with the GDELT/media-market measure.

Revisiting Survey Sources: Pew ATP and ANES

Two established public-opinion sources were evaluated as a completeness check on the news-based approach: the Pew Research Center American Trends Panel^[18] (ATP), a probability-based national panel of roughly 10,000 U.S. adults with dedicated 2024-cycle waves on immigration, abortion and cultural issues, gender identity, and top-problems-facing-the-country — a strong topical match to the four GDELT issues — and the American National Election Studies^[19] (ANES) 2024 Time Series Study, the discipline’s longest-running election survey, with 5,521 pre-election completions and detailed items on abortion, immigration, the economy, and several other topics overlapping the project’s issue set.

Both hit the same geographic-resolution limit that already shaped the news-based approach, just at different tiers. ATP’s public microdata carries only Census region, Census division, and a metro/non-metro flag — no state variable at all, so Pennsylvania respondents cannot even be isolated. ANES’s public file does include a state identifier, so Pennsylvania respondents can be isolated, but county FIPS code, county name, and Census tract are all redacted in the public release; obtaining them requires the same kind of ICPSR restricted-use agreement already ruled out for timeline reasons above. Even with restricted access granted, ANES’s roughly 5,500 national respondents would yield only about 100–250 Pennsylvania cases — thin for a single state-level estimate and far too few to split meaningfully across 67 counties (Table 8).

	Pew ATP	ANES 2024
Topic overlap with GDELT issue set	Strong	Strong
State identifier (public release)	No	Yes

	Pew ATP	ANES 2024
County identifier	No, at any access tier	Restricted-use only, via ICPSR
Access effort	Low — free account, instant download	Low (public file) / high (restricted geography, multi-week ICPSR process)
Usable Pennsylvania sample size	Unknown, likely small	~100–250 statewide; thinner still per county

Table 8. Pew ATP versus ANES 2024 on the dimensions relevant to a Pennsylvania county-level text/sentiment measure.

Neither source can plug into the county/media-market pipeline as a like-for-like replacement for the GDELT measure: ATP blocks at the state level, ANES blocks at the county level behind a multi-week access process. Their more realistic use is as an external check rather than a new model input — comparing the direction of demographic effects already found in the Ridge model against parallel patterns in ATP/ANES topline, or citing specific ATP/ANES findings to justify why crime, immigration, abortion, and the economy were the four issues selected for the GDELT sentiment pull in the first place.

Taken together, this screening process pointed to the same conclusion from two directions: the free, immediately accessible GDELT news archive, aggregated to Pennsylvania’s 11 media markets to resolve the county-level sparsity problem, was the most defensible text/sentiment measure available on this project’s timeline. The construction and results of that measure are described next.

Text-Based Sentiment Extension

The demographics-only model above left roughly half of the out-of-sample variance in Democratic vote-share change unexplained (Table 6). This subsection tests whether county-level news sentiment on four salient campaign issues — crime, immigration, abortion, and the economy — closes any of that remaining gap.

Text Data Construction

Most of Pennsylvania’s 67 counties receive too little dedicated local news coverage to support a reliable county-level text measure on their own, so coverage was aggregated at the media-market level instead. Each county was mapped to its Nielsen designated market area (DMA) using a published county-to-DMA crosswalk^[16], collapsing the 67 counties into 11 media markets spanning Pennsylvania and its border regions. Average news tone on each of the four issues was then pulled for each market from the GDELT Project’s public news archive^[17], restricted to the pre-election window (January–November 2024).

Coverage volume varied sharply by market: the seven markets holding most of Pennsylvania’s population (Philadelphia, Pittsburgh, Harrisburg–Lancaster–York, Wilkes-Barre–Scranton, Erie, Johnstown–Altoona, and the Washington, DC–adjacent market) had hundreds to tens of thousands of articles per issue, while four small markets covering only a sliver of the state had too few articles to trust. Market/issue combinations with fewer than 15 articles were treated as unreliable and replaced with the cross-market median; this affected 5 of the 67 counties, each flagged with an indicator variable so the imputation is traceable rather than silent.

The four resulting tone measures were added to the six demographic predictors from the model above, for a total of 10 predictors, and put through the same pipeline: a VIF diagnostic on the raw predictors, standardization, and RidgeCV with leave-one-out cross-validation over the same 100-point log-spaced alpha grid (0.01 to 10,000).

Extended Model Results

Table 9 reports the VIF for all 10 raw predictors. The four tone measures carry elevated VIFs — immigration tone (246.7) and crime tone (144.9) are both highly collinear with the rest of the predictor set — which is expected given that every county sharing a media market is assigned an identical tone score, so tone tracks whatever demographic and urban/rural composition already distinguishes that market’s counties.

Variable	VIF
Median household income	287.04
Immigration coverage tone	246.71
White population share	186.96
Crime coverage tone	144.89
Abortion coverage tone	81.26
Poverty rate	58.64
Economy coverage tone	46.43
Bachelor’s degree+	41.23
Unemployment rate	21.29
Foreign-born share	17.78

Table 9. Variance inflation factors for the 10-variable model (six demographic predictors plus four news-sentiment measures).

Leave-one-out cross-validation selected a larger penalty value than the demographics-only model ($\alpha = 18.738$ versus 9.326), consistent with the added collinearity from the tone measures. Figure 9 shows all 10 standardized coefficients ranked by magnitude. Unemployment rate and foreign-born share remain the two strongest predictors by a wide margin, retaining the same negative sign as in the demographics-only model; all four sentiment variables rank in the bottom half by magnitude, with abortion coverage tone the weakest predictor in the model.

Demographic predictors still dominate after adding news-sentiment measures

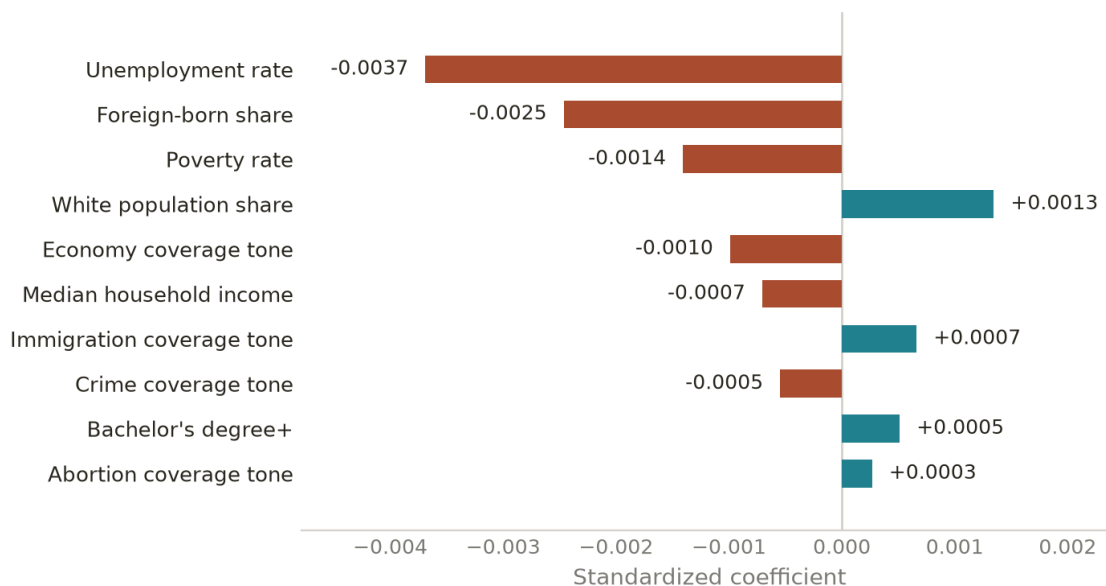


Figure 9. Standardized Ridge regression coefficients, 10-variable model (demographics plus news-sentiment tone). Bars show direction and relative magnitude; all predictors are z-scored, so magnitudes are directly comparable.

Table 10 compares fit statistics against the demographics-only model. In-sample R^2 is essentially flat (0.569 to 0.563), but the honest, out-of-sample LOOCV R^2 ticks down from 0.479 to 0.443 — adding the four sentiment variables did not improve, and slightly worsened, the model’s ability to predict a county’s vote-share change from data it had not seen.

Metric	Demographics only	+ Text/sentiment
Selected alpha (lambda)	9.326	18.738
In-sample R^2	0.569	0.563
LOOCV R^2 (out-of-sample)	0.479	0.443
LOOCV RMSE	0.00721	0.00746

Table 10. Ridge model fit statistics, demographics-only versus 10-variable (+ text/sentiment) model. The out-of-sample LOOCV R^2 row is the primary comparison.

Discussion

Adding county-level news sentiment did not close the gap left by the demographics-only model — if anything, the honest out-of-sample fit moved slightly backward. This is a legitimate, reportable null result for H3’s text-based extension, but it should be read as evidence against this particular operationalization of news sentiment rather than as evidence that local media narrative is irrelevant to Pennsylvania’s vote-share shift.

The most likely explanation is methodological rather than substantive. Media-market aggregation means every county sharing a DMA receives an identical tone score, so the measure captures broad regional campaign narrative rather than genuine county-level variation — and much of what GDELT indexes as “local” coverage of a place name is often

national wire content rather than dedicated local reporting. Both effects would tend to wash out real signal well before it reached the Ridge model, independent of whether issue-based sentiment actually matters to voters.

The core demographic result is, if anything, reinforced by this extension: unemployment rate and foreign-born share remain the two dominant predictors with unchanged signs even after adding four more variables and a larger penalty, and the demographics-only LOOCV R^2 of 0.479 stands as the more defensible estimate of how much of Pennsylvania's county-level Democratic decline is explained by measurable structural factors. A finer-grained, genuinely county-level text measure — rather than one shared across an entire media market — would be needed to test the sentiment hypothesis more conclusively.

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